

Dr. Babasaheb Ambedkar Technological University, Lonere

Regular and Supplementary Winter Examination – 2023

Course: B. Tech.

Branch : Civil Engineering

Semester :V

Subject Code & Name: BTCVC502 Geotechnical Engineering

Max Marks: 60

Date: 03/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Mark
Q. 1 Solve <u>any two</u> of the following.		12
A) Explain the Following terms: A.1) Voids Ratio A.2) Porosity A.3) Degree of saturation A.4) Water Content A.5) Specific Gravity A.6) Air Content	Understand/ CO1	6
B) Why soil is considered as three phase system? Explain with neat sketch?	Understand/ CO1	6
C) State and explain the different types of soil structures?	Understand/ CO1	6
Q.2 Solve <u>any two</u> of the following.		12
A) Enlist the different soil classification systems that were developed by various organizations for their specific purpose. Explain in detail the by 'Indian Soil Classification system'.	CO2/ Apply	6
B) A soil sample of saturated clay has a diameter of 50 mm and the height of 100 mm. The mass of saturated sample is 220 g and its mass when oven dried is 150 g. Find: 1) Water content of the clay & 2) Void Ratio	Understand / CO1	6
C) A soil sample has wet density 20 kN/m^3 and dry density of 18 kN/m^3 . If the specific gravity of soil is 2.67. Calculate the porosity, moisture content and degree of saturation. Assume unit weight of water = 10 kN/m^3	Apply /CO2	6
Q. 3 Solve <u>any two</u> of the following.		12
A) What is mean by permeability of soil. Explain in detail the concept of Hydraulic gradient and Darcy Law?	Understand / CO2	6
B) Determine the average coefficient of permeability in the horizontal and vertical directions for a deposit consisting of three layers of thickness 5 m, 1 m and 2.5 m and having the coefficients of permeability of $3 \times 10^{-2} \text{ mm/sec}$, $3 \times 10^{-5} \text{ mm/sec}$, and $4 \times 10^{-2} \text{ mm/sec}$, respectively. Assume the layers are isotropic.	Apply /CO3	6
C) In falling head permeability test, the head causing flow was initially 90 cm, and it drops 6 cm in 15 minutes. How much time is required for the head to fall to 45 cm.	Apply /CO3	6

Q.4 Solve any two of the following. **12**

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| A) Explain the fundamental equation of shear strength of soil given by Coulomb. | Understand
/ CO2 | 6 |
| B) Explain the different types of shear test based on drainage conditions. | Apply
/ CO3 | 6 |
| C) A vane 100 mm in diameter and 200 mm in heights was pressed into soft clay in a borehole. The torque was applied and gradually increased to 100 N-m when failure took place. Determine the undrain shear strength. | Understand
/ CO2 | 6 |

Q. 5 Solve any two of the following. **12**

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| A) Explain in detail the major factors which affect the compaction? | Understand/
CO2 | 6 |
| B) An earth embankment is compacted at a water content of 12 % to a bulk density of 25 kN/m ³ . If the specific gravity of the sand is 2.65, find the porosity, void ratio and the degree of saturation of the compacted embankment. | Understand
/ CO2 | 6 |
| C) A gravity retaining wall retains 10 m of a backfill, $\gamma = 25 \text{ kN/m}^3$ $\phi = 20^\circ$ with a uniform horizontal surface. Determine the total active thrust per meter run of wall. Assume the wall interface to be vertical. | Apply
/ CO3 | 6 |

***** End *****